

Unconditional Velocity-Defect Bound and the Reduction of Bad-Level Kinetic to a Homothetic-Trace Question

Abstract

We prove an unconditional pointwise inequality

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq \left(\frac{P(B_\rho)}{P(E_\rho)} \right)^2 D_H(t(\rho)) \leq D_H(t(\rho))$$

for every a.e. finite-perimeter torsion level $\rho \in [\rho_*, 1]$. The bound holds on *all* levels, not only the Fusco–Julin good set, and the prefactor $(P(B_\rho)/P(E_\rho))^2 \leq 1$ is itself small on isoperimetrically-bad levels. Integrating against $d\mu = -t' d\rho$ and using the level-set deficit identity gives, with no further hypothesis,

$$\int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C(n, \rho_*) \delta_T(\Omega).$$

The remaining obstruction to closing the bad-level kinetic action $\int_{B_I} |\dot{F}_\rho|_\chi^2 d\mu \leq C\delta$ is therefore reduced to the analogous integrated bound for the *homothetic* term $\int |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1}$ for some Borel centre field z_ρ defined on bad levels. We sketch two routes for this remaining piece — the centroid space–time Π -identity and a direct L^1 -radial-trace construction — and identify which inputs they need.

1 Setup

Adopt the notation of `WIP.WeightedMetricTrace.tex`: torsion function $u = u_\Omega$, $-\Delta u = 1$, levels $E_\rho = \{u > t(\rho)\}$, $|E_\rho| = \omega_n \rho^n$, $w(\rho) = -t'(\rho) \geq 0$, $d\mu = w d\rho$, $V_\rho = w/|\nabla u|$, $\bar{V}_\rho = P(B_\rho)/P(E_\rho)$, D_H the Cauchy–Schwarz defect $D_H = \alpha\beta - P^2$ with $\alpha = \int 1/|\nabla u| d\mathcal{H}^{n-1}$, $\beta = \int |\nabla u| d\mathcal{H}^{n-1} = |E_\rho|$, $P = P(E_\rho)$, $P(B_\rho) = n\omega_n \rho^{n-1}$.

For convenience write $P^* := P(B_\rho)$. The fundamental flux identity $w\alpha = P^*$ (consequence of $|E_\rho| = \omega_n \rho^n$ and $\alpha = -m'(t(\rho))$, $w = -t'(\rho)$) is used throughout.

2 The unconditional velocity-defect bound

Lemma 2.1 (Weighted variance of V). *For a.e. $\rho \in [\rho_*, 1]$ at which E_ρ has finite perimeter,*

$$\int_{\partial^* E_\rho} \frac{(V_\rho - \bar{V}_\rho)^2}{V_\rho} d\mathcal{H}^{n-1} = \frac{P^*}{P^2} D_H(t(\rho)). \quad (1)$$

Proof. Set $f := V_\rho = w/|\nabla u|$, $\bar{f} := \bar{V}_\rho = P^*/P$. Expanding $(f - \bar{f})^2/f = f - 2\bar{f} + \bar{f}^2/f$ and integrating,

$$\int (f - \bar{f})^2/f = \int f - 2\bar{f}P + \bar{f}^2 \int 1/f.$$

Now $\int f = \int V_\rho = w\alpha = P^*$ (flux); $\int 1/f = \int |\nabla u|/w = \beta/w$; $\bar{f}P = P^*$ and $\bar{f}^2 = P^{*2}/P^2$. Hence

$$\int (f - \bar{f})^2/f = P^* - 2P^* + \frac{P^{*2}}{P^2} \cdot \frac{\beta}{w} = -P^* + \frac{P^*}{P^2} \cdot \frac{P^*\beta}{w} = -P^* + \frac{P^*}{P^2} \alpha\beta,$$

where the last equality used $w\alpha = P^*$ to write $P^*\beta/w = \alpha\beta$. Therefore

$$\int (f - \bar{f})^2/f = \frac{P^*}{P^2} (\alpha\beta - P^2) = \frac{P^*}{P^2} D_H. \quad \square$$

Theorem 2.2 (Unconditional L^1 velocity-defect bound). *For a.e. $\rho \in [\rho_*, 1]$ at which E_ρ has finite perimeter,*

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq \left(\frac{P^*}{P} \right)^2 D_H(t(\rho)) \leq D_H(t(\rho)). \quad (2)$$

No isoperimetric smallness or Fusco–Julin good-level assumption is used.

Proof. By Cauchy–Schwarz in the form $(\int |g| d\mathcal{H}^{n-1})^2 \leq (\int g^2/V_\rho d\mathcal{H}^{n-1})(\int V_\rho d\mathcal{H}^{n-1})$ applied to $g := V_\rho - \bar{V}_\rho$ (with the standard $\sqrt{V_\rho} \cdot \sqrt{1/V_\rho}$ -factorisation),

$$\left(\int |V_\rho - \bar{V}_\rho| \right)^2 \leq \left(\int \frac{(V_\rho - \bar{V}_\rho)^2}{V_\rho} \right) \left(\int V_\rho \right) = \frac{P^*}{P^2} D_H \cdot P^* = \left(\frac{P^*}{P} \right)^2 D_H,$$

using Lemma ?? and $\int V_\rho = P^*$. The second inequality of (2) follows from $P \geq P^*$ (isoperimetric inequality). $\square$

Remark 2.3 (Strengthening over `TRIMMEDVELOCITYREPAIR`). The estimate (2) is sharper than the bound proved in `WIP_TrimmedVelocityRepair.tex` in two ways.

1. It is *unconditional*: the previous argument restricted to the Fusco–Julin good set $G_I = \{D_I \leq \eta_I\}$ via a truncation $\{f \geq \bar{f}/2\}$. No such restriction is needed here. The Cauchy–Schwarz $(\int |g|)^2 \leq (\int g^2/V)(\int V)$ does not require a lower bound on V : it weights the variance by V itself, making the integrand integrable wherever V is.
2. The prefactor $(P^*/P)^2 \leq 1$ is itself *smaller* on isoperimetrically bad levels. Since $\mathcal{I}(\rho) = P^2 - P^{*2} = (P - P^*)(P + P^*)$, we have $(P^*/P)^2 = 1 - \mathcal{I}/P^2$. In the regime where $P \gg P^*$ on a bad level, the velocity defect is suppressed by a factor $\leq (P^*/P)^2 \leq P^{*2}/(P^{*2} + \mathcal{I})$.

Corollary 2.4 (Integrated velocity-defect on all levels).

$$\int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq \int_{\rho_*}^{\rho_\delta} D_H(t(\rho)) d\mu(\rho) \leq C(n, \rho_*) \delta_T(\Omega). \quad (3)$$

Proof. Apply Theorem 2.2 pointwise, then integrate; the second inequality is the level-set deficit identity of `WIP_LevelSetIdentity.tex`. $\square$

3 Reduction of the bad-level kinetic problem

Combining (2) with the unconditional metric upper gradient of `WIP_LimsupDeformation.tex`,

$$|\dot{F}_\rho|_{\mathcal{X}} \leq \rho^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_{\rho, \rho}} - a \cdot \nu_\rho| d\mathcal{H}^{n-1},$$

specialising to $a = 0$, and the triangle inequality $|V_\rho - H_{z_\rho, \rho}| \leq |V_\rho - \bar{V}_\rho| + |H_{z_\rho, \rho} - \bar{V}_\rho|$, we obtain pointwise

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq 2\rho^{-2n} \left[D_H(t(\rho)) + \left(\int |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \right] \quad (4)$$

for any bounded Borel centre field z_ρ . This holds on *every* a.e. finite-perimeter level, with no FJ smallness.

Proposition 3.1 (Reduction to homothetic-trace). *Suppose there is a bounded Borel centre field $\rho \mapsto z_\rho \in \mathbb{R}^n$, $|z_\rho| \leq R'$, and a constant $C_{\text{hom}} = C_{\text{hom}}(n, R, R', \rho_*) > 0$ such that*

$$\int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C_{\text{hom}} \delta_T(\Omega). \quad (5)$$

Then

$$\int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\mu(\rho) \leq C(n, R, R', \rho_*) \delta_T(\Omega). \quad (6)$$

In particular both bad-level kinetic hypotheses (eq:bad-I-action) and (eq:bad-tau-action) of *WIP_WeightedMetricTrace.tex* are discharged.

Proof. Integrate (4) against $d\mu$ and use Corollary 2.4 for the first piece, (5) for the second, and $\rho^{-2n} \leq \rho_*^{-2n}$ on $[\rho_*, \rho_\delta]$. The bad- τ kinetic follows by the standard $w \geq \tau_0$ -on- G_τ split applied to (6) inside *WIP_WeightedMetricTrace.tex*'s Theorem 3.2 proof, and the bad- I kinetic is a direct corollary because B_I has finite μ -measure inside $[\rho_*, \rho_\delta]$. $\square$

4 Status of (5): two candidate routes

The conditional input (5) is the *only* remaining input needed to make Plan 2 / Route δ unconditional. Two routes for proving it are visible.

4.1 Route I: centroid space–time Π -identity

Choose $z_\rho := \bar{z}_\rho$, the centroid of E_ρ . The divergence identity [4, Eq. (5.3')],

$$\int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{z_\rho}|^2 d\mathcal{H}^{n-1} = 2(P(E_\rho) - P(B_\rho)) + 2\Pi(E_\rho, \bar{z}_\rho),$$

combined with the radial trace estimate of [5, Thm. 2.2], gives

$$\left(\int |H_{\bar{z}_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_*) \left[|E_\rho \Delta B_\rho(\bar{z}_\rho)|^2 + \left(\int |\nu - e_{z_\rho}|^2 d\mathcal{H}^{n-1} \right)^2 \right]. \quad (7)$$

By the pointwise slice-rearrangement inequality $|E_\rho \Delta B_\rho(\bar{z}_\rho)|^2 \leq C \Pi(E_\rho, \bar{z}_\rho)$ of [4, Lem. 4.3] and the integrated identity

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu \leq C_\Pi \delta_T \quad (\Pi\text{-int})$$

of the same block, *plus* an integrated bound on $\left(\int |\nu - e_{z_\rho}|^2 \right)^2$ which is open (naive Cauchy–Schwarz only yields a $\sqrt{\delta_T}$ rate, not δ_T), one obtains (5). The remaining sub-target inside Route I is therefore

$$\int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |\nu - e_{z_\rho}|^2 d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C \delta_T,$$

which is currently isolated as a separate open problem.

4.2 Route II: oriented radial flux on bad levels

The oriented divergence trace [5, Thm. 2.1] gives, for *any* centre z with $|z| \leq R + 1$,

$$\int_{\partial^* E_\rho} \left| \frac{|x-z|}{\rho} - 1 \right| d\mathcal{H}^{n-1} \leq \frac{n}{\rho_*} |E_\rho \Delta B_\rho(z)| + \frac{W(R, \rho_*)}{2} \int |\nu - e_z|^2 d\mathcal{H}^{n-1},$$

unconditionally on the deficit. If z is chosen as the *Fraenkel* centre $z_F = z_F(E_\rho)$, then $|E_\rho \Delta B_\rho(z_F)|^2 \leq C_F \mathcal{I}(\rho)$ holds unconditionally (Figalli–Maggi–Pratelli). The squared integrated $\int |\nu - e_{z_F}|^2 d\mathcal{H}^{n-1}$ at the *Fraenkel* centre is not $\leq C\mathcal{I}$ in general — that is the input that fails when one tries to use the Fraenkel centre as a substitute for the FJ centre on bad levels. Closing Route II therefore reduces to bounding $\int (\int |\nu - e_{z_F}|^2)^2 d\mu$ separately, with the same kind of structural difficulty as Route I.

5 Summary of progress

- Theorem 2.2 is new: the L^1 -velocity defect $\int |V - \bar{V}| d\mathcal{H}^{n-1}$ is controlled by $\sqrt{D_H}$ on *every* finite-perimeter level, with prefactor $P^*/P \leq 1$. The previous TRIMMEDVELOCITYREPAIR bound was good-level only.
- Corollary 2.4 integrates this against $d\mu$ unconditionally, giving the half of the kinetic action that involves D_H directly.
- Proposition 3.1 reduces the bad-level kinetic problem to a single integrated homothetic-trace bound (5).
- Section 4 isolates the residual difficulty as

$$\int_{\rho_*}^{\rho_\delta} \left(\int |\nu - e_{z_\rho}|^2 d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C \delta_T,$$

for either the centroid centre $z_\rho = \bar{z}_\rho$ (Route I, via II-int) or the Fraenkel centre $z_\rho = z_F$ (Route II, direct). This is a clean, decoupled mathematical question and not specific to Fusco–Julin.

References

- [1] Plan 2 building block LEVELSETIDENTITY, WIP_LevelSetIdentity.tex.
- [2] Plan 2 Deliverable A, LIMSUPDEFORMATION, WIP_LimsupDeformation.tex.
- [3] Plan 2 building block TRIMMEDVELOCITYREPAIR, WIP_TrimmedVelocityRepair.tex.
- [4] Plan 2 auxiliary block SPACETIMEIDENTITY, WIP_SpaceTimeIdentity.tex.
- [5] Plan 2 bridge block FJCENTERBRIDGE, WIP_FJCenterBridge.tex.
- [6] Plan 2 building block WEIGHTEDMETRICTRACE, WIP_WeightedMetricTrace.tex.